WORKED SOLUTION

STEP III 1992, Question 10

Pure • Arc Length and Area

STEP III 1992, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q10

The key idea

Both perimeters are arc-length integrals that can't be done in elementary terms (they are *elliptic integrals*), so the only way to show they are equal is to turn one integral into the other. Write down the arc-length element for each curve, simplify each integrand using $\cos^2 + \sin^2 = 1$, and then a substitution and a little symmetry show that the two are the same integral. The areas are easier because both can be evaluated directly.

(The range printed in the paper ends with $\frac{1}{4}\theta$. This is clearly a misprint for $\frac{1}{4}\pi$, and we take $-\frac{1}{4}\pi < \theta < \frac{1}{4}\pi$.)

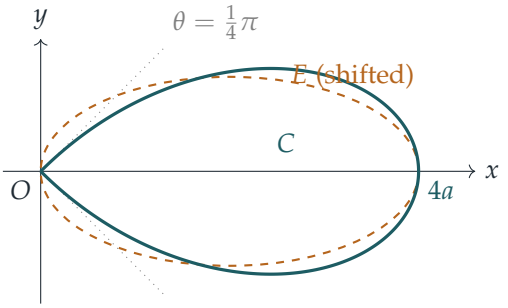
Solution

Sketching C

For $-\frac{1}{4}\pi < \theta < \frac{1}{4}\pi$ we have $-\frac{1}{2}\pi < 2\theta < \frac{1}{2}\pi$, so $r = 4a\cos 2\theta > 0$ throughout (taking $a > 0$). Some features:

- $r(-\theta) = r(\theta)$, so C is symmetric about the initial line.
- The maximum $r = 4a$ is at $\theta = 0$, the point $(4a, 0)$.
- As $\theta \to \pm\frac{1}{4}\pi$, $r \to 0$, so the curve runs into the origin, tangent there to the lines $\theta = \pm\frac{1}{4}\pi$.

So C is a single closed loop, one petal of the four-petalled rose $r = 4a\cos 2\theta$, lying along the positive x-axis. The figure shows it together with a copy of E translated by $2a$ so that it spans the same interval $0 \le x \le 4a$.



Perimeter of C

For a polar curve the element of arc length satisfies $\left(\frac{\mathrm{d}s}{\mathrm{d}\theta}\right)^2 = r^2 + \left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2$. Here $\frac{\mathrm{d}r}{\mathrm{d}\theta} = -8a\sin 2\theta$, so

$$\begin{aligned} r^2 + \left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2 &= 16a^2\cos^2 2\theta + 64a^2\sin^2 2\theta \\ &= 16a^2\left(1 + 3\sin^2 2\theta\right). \end{aligned}$$

Hence, since the loop is traced once as θ runs over $(-\frac{1}{4}\pi, \frac{1}{4}\pi)$,

$$\text{perimeter of } C = \int_{-\pi/4}^{\pi/4} 4a\sqrt{1 + 3\sin^2 2\theta}\,\mathrm{d}\theta.$$

Substitute $\phi = 2\theta$, so $\mathrm{d}\theta = \frac{1}{2}\mathrm{d}\phi$ and the limits become $\pm\frac{1}{2}\pi$:

$$\text{perimeter of } C = 2a\int_{-\pi/2}^{\pi/2}\sqrt{1 + 3\sin^2\phi}\,\mathrm{d}\phi = 4a\int_0^{\pi/2}\sqrt{1 + 3\sin^2\phi}\,\mathrm{d}\phi,$$

the last step because the integrand is an even function of ϕ.

Perimeter of E

With $x = 2a\cos\phi$, $y = a\sin\phi$,

$$\left(\frac{\mathrm{d}x}{\mathrm{d}\phi}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}\phi}\right)^2 = 4a^2\sin^2\phi + a^2\cos^2\phi = a^2\left(1 + 3\sin^2\phi\right),$$

and E is traced once as ϕ runs from 0 to 2π, so

$$\text{perimeter of } E = a\int_0^{2\pi}\sqrt{1 + 3\sin^2\phi}\,\mathrm{d}\phi.$$

? **Ask yourself** Why is $\int_0^{2\pi}\sqrt{1 + 3\sin^2\phi}\,\mathrm{d}\phi$ four times the integral over $[0, \frac{1}{2}\pi]$?

Write $g(\phi) = \sqrt{1 + 3\sin^2\phi}$. Because $\sin^2(\pi - \phi) = \sin^2\phi$, the graph of g on $[\frac{1}{2}\pi, \pi]$ is the mirror image of its graph on $[0, \frac{1}{2}\pi]$, so $\int_0^{\pi} g = 2\int_0^{\pi/2} g$. Because $\sin^2(\phi + \pi) = \sin^2\phi$, g has period π, so $\int_{\pi}^{2\pi} g = \int_0^{\pi} g$. (Geometrically, the four quadrants of the ellipse have equal length.) Therefore

$$\text{perimeter of } E = 4a\int_0^{\pi/2} \sqrt{1 + 3\sin^2\phi}\,\mathrm{d}\phi,$$

which is exactly the expression we found for the perimeter of C. So the two perimeters are equal, and we never had to evaluate the integral.

Areas

The area enclosed by a polar curve is $\frac{1}{2}\int r^2\,\mathrm{d}\theta$. Using $\cos^2 2\theta = \frac{1}{2}(1 + \cos 4\theta)$,

$$\begin{aligned}\text{area inside } C &= \frac{1}{2}\int_{-\pi/4}^{\pi/4} 16a^2\cos^2 2\theta\,\mathrm{d}\theta = 4a^2\int_{-\pi/4}^{\pi/4}(1 + \cos 4\theta)\,\mathrm{d}\theta \\ &= 4a^2\Big[\theta + \tfrac{1}{4}\sin 4\theta\Big]_{-\pi/4}^{\pi/4} = 4a^2\cdot\frac{\pi}{2} = 2\pi a^2.\end{aligned}$$

For the ellipse, the area enclosed by a closed curve traced anticlockwise is $\frac{1}{2}\oint\left(x\frac{\mathrm{d}y}{\mathrm{d}\phi} - y\frac{\mathrm{d}x}{\mathrm{d}\phi}\right)\mathrm{d}\phi$, which gives

$$\text{area inside } E = \frac{1}{2}\int_0^{2\pi}\left(2a^2\cos^2\phi + 2a^2\sin^2\phi\right)\mathrm{d}\phi = \frac{1}{2}\cdot 2a^2\cdot 2\pi = 2\pi a^2.$$

(Equally, this is the standard $\pi \times$ semi-axis $\times$ semi-axis $= \pi(2a)(a)$.) So the two areas are equal, both being $2\pi a^2$.

Checks

- With $a = 1.7$, approximating each curve by a polygon of 200 000 sides gave both perimeters as 16.470362, which equals $4a\int_0^{\pi/2}\sqrt{1 + 3\sin^2\phi}\,\mathrm{d}\phi$ by Simpson's rule. So each perimeter is about $9.6884a$.
- The same polygons gave both areas as $18.158406 = 2\pi(1.7)^2$.
- A sanity check on the area of C: it lies inside the rectangle $0 \le x \le 4a$, $|y| \le 1.1a$, which has area about $8.8a^2$, and $2\pi a^2 \approx 6.3a^2$ is a sensible fraction of that.

Common slips

- Forgetting the factor $\frac{1}{2}$ from $\mathrm{d}\theta = \frac{1}{2}\mathrm{d}\phi$ when substituting $\phi = 2\theta$. The factor $4a$ for C becomes $2a$, and only becomes $4a$ again after the even-function step.
- Using $\int r\,\mathrm{d}\theta$ instead of $\frac{1}{2}\int r^2\,\mathrm{d}\theta$ for the polar area.
- Treating the whole rose $r = 4a\cos 2\theta$ for $0 \le \theta < 2\pi$ as C. The given range picks out a single petal.

Beyond STEP

The integral $\int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \phi}\, \mathrm{d}\phi$ is the *complete elliptic integral of the second kind*, and it has no elementary antiderivative. That's why the question forbids evaluating the integrals: the perimeter of an ellipse has no closed form. Our integral is the case $k^2 = -3$. It can be rewritten in the standard form with $0 < k < 1$ by replacing $\sin^2 \phi$ with $1 - \cos^2 \phi$ and then ϕ with $\frac{1}{2}\pi - \phi$. Elliptic integrals led to elliptic functions and elliptic curves, which are central to modern number theory. Siklos's *Advanced Problems in Mathematics* has more STEP questions where arc lengths and areas of polar and parametric curves are compared like this.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.